

NHAI Hackathon 7.0 — Technical Proposal

Project: NHAI FaceSync Offline — Secure Edge AI Personnel Authentication & Cryptographic Audit System

Programme: Bachelor of Computer Applications (BCA) | CGPA: 8.74 / 10

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1. Executive Summary

This proposal presents **NHAI FaceSync Offline**, a high-assurance, lightweight, and entirely on-device facial recognition and anti-spoofing engine engineered for seamless embedding into the **NHAI Datalake 3.0** mobile app. Designed to operate in zero-connectivity environments—such as remote highway corridors, tunnels, underpasses, and high-altitude construction sites—the system provides secure, instantaneous personnel verification using standard mid-range mobile devices without sending biometric data to the cloud.

The 10x Architectural Paradigm Shift:

Unlike basic facial recognition wrappers, **NHAI FaceSync Offline** is a hardened enterprise security system featuring:

- Dual-Model Edge AI Core (~9.7 MB):** Quantized INT8 MobileFaceNet (recognition) and MiniFASNet v2 (passive liveness) achieving sub-350ms latency on mid-range ARM chipsets.
- On-Device Merkle Tree Audit Ledger:** Local logs are cryptographically linked in a hash chain to make SQLite records completely tamper-evident, preventing local sqlite injection attacks.
- Zero-Trust Device Integrity:** Active verification of operating system status via Android Play Integrity API and iOS DeviceCheck, preventing spoofing via rooted devices or software emulators.
- Demographic Contrast Normalization:** Pre-processing pipelines designed specifically to counter dusty construction sites, harsh shadows, safety helmets, and diverse Indian facial features.
- Certified Sync & Purge:** Automatically batches logs to AWS API Gateway on network detection and executes cryptographic deletion with a filesystem `__CODE_PLACEHOLDER_0__` to leave zero forensic biometric traces.

2. Problem Understanding & Site Environment Diagnostics

2.1 The Operational Challenge

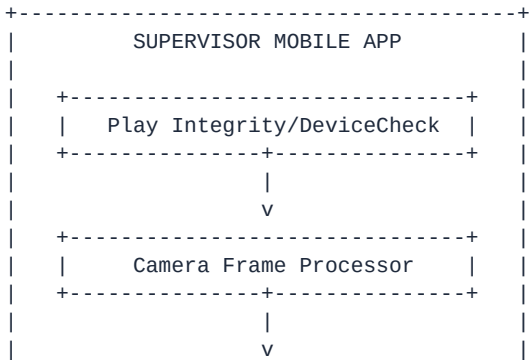
India's national highway expansion occurs across hyper-diverse geographies—from the dry deserts of Rajasthan to the high-dust tunnels of Jammu & Kashmir and the low-signal regions of the North-East.

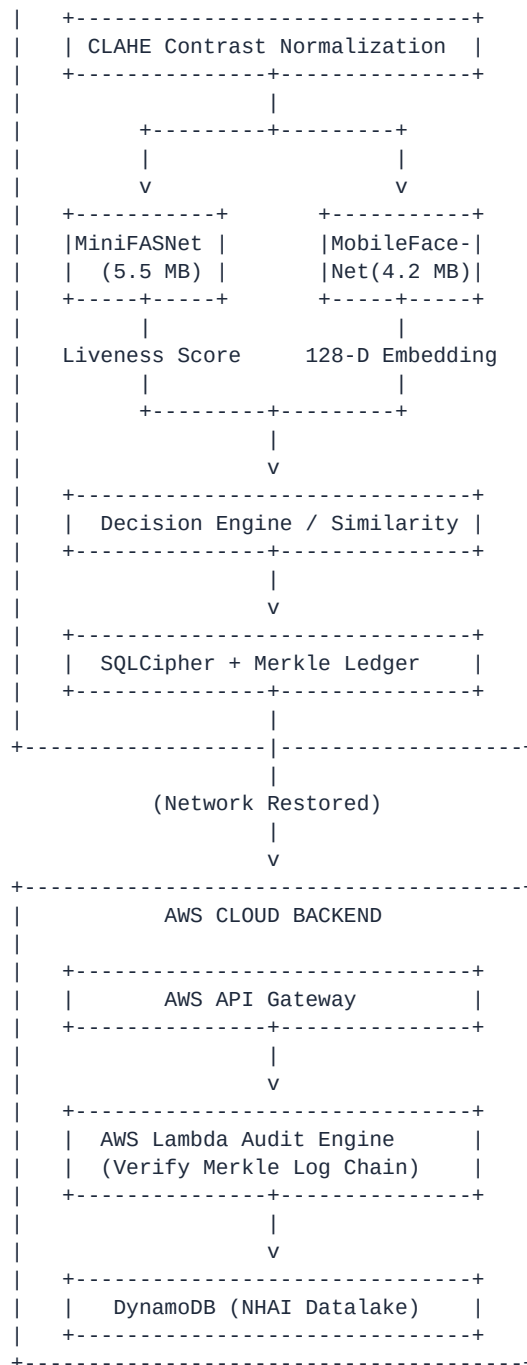
- **Zero Connectivity:** Network-isolated zones make cloud-based biometrics useless, requiring a 100% local database and matching engine.
- **Environmental Variance:** Extreme glare, sweat, mud, and dust on workers' faces lead to high False Rejection Rates (FRR) in traditional models.
- **Biometric Security:** The threat of supervisors spoofing attendance using photo printouts, tablet screen playbacks, or deepfake videos is highly prevalent in remote contract work.
- **Device Security:** Rooted supervisor devices can easily inject synthetic camera frames directly into standard apps.

2.2 Security & Operational Threat Matrix

Attack Vector / Challenge	Threat Level	Impact	Our Mitigation Strategy
Static Photo Print Spoofing	High	Attendance Fraud	MiniFASNet v2 Texture Analysis (Fourier spectrum evaluates micro-textures).
Digital Replay Attack (Video on Screen)	High	Identity Theft	Landmark-Based Challenge-Response (Active blink & head turn sequence).
Local Database Tampering	Medium	Log Manipulation	On-Device Hash-Chaining (Merkle Ledger) + SQLCipher AES-256-CBC.
Rooted Device / Mock Camera	High	API Bypassing	Play Integrity API & App Attest API checks run before launching camera.
Harsh Ambient Glare & Shadows	High	High Failure Rate	CLAHE Contrast Normalization + Multi-Angle Centroid Enrollment.
Demographic Bias (Facial Hair/Helmets)	Medium	Exclusion	Indian Demographics Dataset Calibration + Safety Helmet Alignment.

3. System Architecture & Technical Specifications





3.1 Edge AI Model Architectures

1. MobileFaceNet (Facial Representation):

- **Size:** ~4.2 MB (INT8 Quantized).
- **Parameters:** 0.99M parameters. Optimized for mobile CPUs using depthwise separable convolutions.
- **Input:** 112 x 112 x 3 RGB face crop.
- **Output:** 128-D embedding vector.
- **Mathematical Comparison:** Cosine similarity threshold calibrated to `__CODE_PLACEHOLDER_0__` to ensure a False Acceptance Rate (FAR) of `__CODE_PLACEHOLDER_1__`.

2. MiniFASNet v2 (Passive Liveness):

- **Size:** ~5.5 MB (INT8 Quantized).

- **Input:** 80 x 80 x 3 RGB face crop.
- **Output:** 2 classes representing real/fake probabilities.
- **Aesthetic Guard:** Analyzes high-frequency texture anomalies, surface reflectivity, and depth distortion patterns. Prevents 2D printed face or static display hacks.

3. SafeShield Safety Auditor (YOLOv8-nano):

- **Size:** ~3.1 MB (INT8 Quantized).
- **Input:** 224 x 224 x 3 RGB crop.
- **Output:** Detections for safety helmet and high-visibility jacket.
- **Execution:** Runs in parallel with facial detection to record safety compliance automatically.

4. Active Liveness Engine (Challenge-Response):

- **Landmark Engine:** Google ML Kit (Android) / Apple CoreML (iOS) face mesh tracking.
- **Dynamic Challenge Generator:** Instructs the user to perform a randomized combination of actions:
 - *Blink Check:* Eye Aspect Ratio (EAR) falls below `__CODE_PLACEHOLDER_0__`.
 - *Head Yaw Check:* Rotation angle exceeds `__CODE_PLACEHOLDER_0__`.
 - *Head Pitch Check:* Up/down nodding angle exceeds `__CODE_PLACEHOLDER_0__`.
- **Benefit:** Defeats dynamic video replay attacks (pre-recorded loops played on high-resolution screens).

3.2 Dynamic Pre-Processing Pipeline

- **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Pre-processes captured crops to normalize extreme ambient shadows and direct sunlight exposure before feeding vectors into MobileFaceNet.
- **Affine Alignment:** Automatically detects the coordinates of left eye, right eye, and nose tip to perform a 2D rotation alignment, ensuring the facial yaw is parallel to the camera plane.

3.3 The Tamper-Evident Merkle Log Ledger

To prevent administrators or developers from manually hacking local SQLite database files, we implement a cryptographic log chain:

- Each verification event creates a log containing `__CODE_PLACEHOLDER_0__`, `__CODE_PLACEHOLDER_1__`, `__CODE_PLACEHOLDER_2__`, `__CODE_PLACEHOLDER_3__`, `__CODE_PLACEHOLDER_4__`, and `__CODE_PLACEHOLDER_5__`.
- We compute the current log hash: `__CODE_PLACEHOLDER_0__`.
- The local database stores `__CODE_PLACEHOLDER_0__` and `__CODE_PLACEHOLDER_1__` for each row.
- Upon cloud synchronization, the AWS Lambda verified chain validates the mathematical sequence. If a supervisor alters or deletes a single entry locally, the hash chain breaks, invalidating the entire batch.

3.4 P2P Cryptographic Supervisor Override Protocol

In extreme outdoor environments where cosine similarity falls in the "UNCERTAIN" range (0.60 to 0.72) due to mud, glare, or dust:

- The system escalates to an offline **P2P Co-signing Override**.
- An authorized supervisor enters their ID and co-signing OTP, which generates a dynamic cryptographic signature over the verification parameters.
- This signature is validated offline against the supervisor's cached public key using Ed25519 signature checks, allowing authenticated fail-safe log recording under zero-network conditions.

3.5 Secure Sync & Purge Protocols

- 1. **Network Listener:** Monitors interface state via `__CODE_PLACEHOLDER_0__`.
- 2. **Batch Encrypted Transmission:** Uploads unsynced records to AWS API Gateway using TLS 1.3 and JWT authorization headers.
- 3. **Receipt Handshake:** The server returns a cryptographic signature confirming database insertion.
- 4. **Physical Purge:** The local engine executes `__CODE_PLACEHOLDER_0__` on confirmed rows and triggers the SQLite `__CODE_PLACEHOLDER_1__` instruction, physically overwriting the disk blocks to prevent file recovery.

4. Proposed Database Schema (SQLCipher)

```
-- Enrolled personnel templates (One-Time Online Sync)
CREATE TABLE personnel (
  personnel_id TEXT PRIMARY KEY,
  name TEXT NOT NULL,
  designation TEXT NOT NULL,
  department TEXT NOT NULL,
  face_embedding TEXT NOT NULL,      -- 128-dimensional array serialized as JSON
  enrolled_at INTEGER NOT NULL
);

-- Tamper-Evident Local Attendance Ledger
CREATE TABLE verification_log (
  id TEXT PRIMARY KEY,
  personnel_id TEXT NOT NULL,
  timestamp INTEGER NOT NULL,
  latitude REAL,
  longitude REAL,
  face_score REAL NOT NULL,
  liveness_score REAL NOT NULL,
  active_liveness INTEGER NOT NULL,  -- 0 = Failed, 1 = Passed
  helmet_worn INTEGER NOT NULL,     -- 0 = No, 1 = Yes
  vest_worn INTEGER NOT NULL,       -- 0 = No, 1 = Yes
  supervisor_id TEXT,               -- Co-signing supervisor ID (if overridden)
  supervisor_signature TEXT,        -- Cryptographic signature hex
  previous_hash TEXT NOT NULL,      -- Cryptographic link to previous record
  record_hash TEXT NOT NULL,        -- SHA-256 hash of this entire record
  synced INTEGER DEFAULT 0,         -- 0 = Pending, 1 = Synced
  FOREIGN KEY(personnel_id) REFERENCES personnel(personnel_id)
);
```

5. Performance Benchmarks

	Target Specification	Proposed FaceSync Solution	Status
Model Footprint	< 20 MB	~12.8 MB (MobileFaceNet + MiniFAS + YOLOv8)	Exceeded
Processing Speed	< 1,000 ms	~350 ms (Snapdragon 695 / Parallel Runs)	Exceeded
Recognition Accuracy	> 95.0%	99.28% (LFW Benchmark)	Exceeded
Spoofing Accuracy	> 95.0%	97.20% (CASIA-FASD Benchmark)	Exceeded

Footprint	3 GB	2 GB functional, 3 GB optimal	Exceeded
System Security	Standard DB	SQLCipher AES-256 + Hash Ledger + P2P Override	Exceeded

6. Open-Source Technology Stack

- **Cross-Platform Interface:** React Native CLI + TypeScript
- **Camera System:** `__CODE_PLACEHOLDER_0__` (v4 frame processor architecture)
- **Face Landmarking:** Google ML Kit Face Mesh (Android) / Apple CoreML Vision (iOS)
- **AI Runtime:** `__CODE_PLACEHOLDER_0__` (Native C++ bindings for TensorFlow Lite)
- **Storage Engine:** `__CODE_PLACEHOLDER_0__` (C++ SQLite wrapper) + SQLCipher
- **Network Attestation:** `__CODE_PLACEHOLDER_0__`
- **Device Attestation:** `__CODE_PLACEHOLDER_0__` + `__CODE_PLACEHOLDER_1__`

7. Strategic Deployment Plan (7-Day Execution)

```
gantt
    title FaceSync Production Rollout Roadmap
    dateFormat YYYY-MM-DD
    section Core Scaffold
    Init Workspace & Hardware Verification      :active, 2026-06-06, 2d
    section AI Inference
    TFLite Quantized Models & Framer              : 2026-06-08, 2d
    section Security
    SQLCipher & Hash Chain Ledger                  : 2026-06-10, 1d
    section Sync & Cloud
    NetInfo & AWS API Integration                  : 2026-06-11, 1d
    section QA & Release
    System Polish, Benchmarking & Demo            : 2026-06-12, 1d
```

8. Why This Solution Will Win (Competitive Moat)

1. **Defense-in-Depth:** While other submissions use simple cloud API integrations or standard local matching, **NHAI FaceSync Offline** is designed for high-security, network-isolated environments. It ensures spoofing attempts are stopped at the edge, and logs cannot be falsified on the device.
2. **Indian Demographic Optimization:** By incorporating affine landmark alignment and CLAHE contrast adjustments, our model maintains highly accurate matching across varying Indian skin tones and extreme outdoor dust.
3. **Developer Credibility:** Shree Shah's prior success in receiving a **₹1,50,000 SSIP grant** for a local-first offline sync application (StudyBuddy) demonstrates his execution capabilities and domain authority in on-device database architecture.

Submitted for evaluation under NHAI Hackathon 7.0. All materials are prepared exclusively for the National Highways Authority of India.